# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

**1.** The domain of the function $\sin^{-1}\left(\log_2\left(\frac{x}{3}\right)\right)$ is -

(A) [1,3] (B) $\left[\frac{1}{2},3\right]$ (C) $\left[\frac{3}{2},6\right]$ (D) $\left[\frac{1}{2},2\right]$

**MIT001**

**2.** Domain of the function $f(x) = \log_e\cos^{-1}\{\sqrt{x}\}$ is, where {.} represents fractional part function –

(A) $x \in \mathbb{R}$ (B) $x \in [0, \infty)$ (C) $x \in (0, \infty)$ (D) $x \in \mathbb{R} - \{x \mid x \in I\}$

**MIT002**

**3.** The range of the function $f(x) = \sin^{-1}(\log_2(-x^2 + 2x + 3))$ is –

(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{2}, 0\right]$ (C) $\left[0, \frac{\pi}{2}\right]$ (D) [–1, 1]

**MIT003**

**4.** Range of $f(x) = \cot^{-1}(\log_e(1 - x^2))$ is -

(A) (0,π) (B) $\left(0,\frac{\pi}{2}\right)$ (C) $\left[\frac{\pi}{2},\pi\right)$ (D) $\left(0,\frac{\pi}{2}\right]$

**MIT004**

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**5.** The function $f(x) = \cot^{-1}\sqrt{(x+3)x} + \cos^{-1}\sqrt{x^2+3x+1}$ is defined on the set $S$, where $S$ is equal to:

(A) {0, 3} (B) (0, 3) (C) {0, –3} (D) [–3, 0]

**MIT005**

**6.** Range of $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$ is

(A) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (B) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (C) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (D) $\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$

**MIT006**

**7.** $\text{cosec}^{-1}(\cos x)$ exists if

(A) $x \in [-1, 1]$ (B) $x \in R$

(C) $x$ is an odd multiple of $\frac{\pi}{2}$ (D) $x$ is a multiple of π

**MIT007**

**8.** The value of $\cot^{-1}\left\{\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}\right\}$, where $\frac{\pi}{2} < x < \pi$, is:

(A) $\pi - \frac{x}{2}$ (B) $\frac{\pi}{2} + \frac{x}{2}$ (C) $\frac{x}{2}$ (D) $2\pi - \frac{x}{2}$

**MIT008**



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**9.** The inequality $\sin^{-1}(\sin 5) > x^2 - 4x$ holds for

(A) $x \in \left(2-\sqrt{9-2\pi}, 2+\sqrt{9-2\pi}\right)$ (B) $x > 2+\sqrt{9-2\pi}$

(C) $x < 2-\sqrt{9-2\pi}$ (D) $x \in \phi$

**MIT009**

**10.** Show that the roots r, s and t of the cubic $x(x-2)(3x-7) = 2$, are real and positive. Also compute the value of $\tan^{-1}(r) + \tan^{-1}(s) + \tan^{-1}(t)$.

(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $2\pi$

**MIT010**

**11.** The complete solution set of the inequality $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \le 0$, where [.] denotes greatest integer function, is

(A) $(-\infty, \cot 3]$ (B) $[\cot 3, \cot 2]$ (C) $[\cot 3, \infty)$ (D) $(-\infty, \cot 2]$

**MIT011**

**12.** If $2 \le a < 3$, then the value of $\cos^{-1}\cos[a] + \text{cosec}^{-1}\text{cosec}[a] + \cot^{-1}\cot[a]$, (where [.] denotes greatest integer less than equal to $x$) is equal to

(A) $2-\pi$ (B) $2+\pi$ (C) $\pi$ (D) 6

**MIT012**

**13.** $\tan^{-1}\left(1-x^2-\frac{1}{x^2}\right)+\sin^{-1}\left(x^2+\frac{1}{x^2}-1\right)$ (where $x \neq 0$) is equal to

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\pi$

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**MIT013**

**14.** If $\cos^{-1}(2x^2-1) = 2\pi - 2\cos^{-1}x$, then -

(A) $x \in [-1, 0]$ (B) $x \in [0, 1]$ (C) $x \in \left[0, \frac{1}{\sqrt{2}}\right]$ (D) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

**MIT014**

**15.** If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, then $4x^2 - 4xy\cos\alpha + y^2$ is equal to-

(A) $2\sin 2\alpha$ (B) 4 (C) $4\sin^2\alpha$ (D) $-4\sin^2\alpha$

**MIT015**

**16.** $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}x\right)$, $x \neq 0$ is equal to

(A) $x$ (B) $2x$ (C) $\frac{2}{x}$ (D) $\frac{x}{2}$

**MIT016**

**17.** If $\alpha = 2\arctan\left(\frac{1+x}{1-x}\right)$ & $\beta = \arcsin\left(\frac{1-x^2}{1+x^2}\right)$, what is the value of $\alpha + \beta$ if $x > 1$.

(A) $-\pi$ (B) $\pi$ (C) $\frac{\pi}{2}$ (D) $2\pi$

**MIT017**



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**18.** The numerical value of $\tan\left(2\tan^{-1}\frac{1}{5}-\frac{\pi}{4}\right)$ is

(A) $\frac{-7}{17}$ (B) $\frac{7}{17}$ (C) $\frac{17}{7}$ (D) $-\frac{2}{3}$

**MIT018**

**19.** $\sum_{r=0}^{\infty}\tan^{-1}\left(\frac{r((r+1)!)}{(r+1)+((r+1)!)^2}\right)$ is equal to -

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\cot^{-1}3$ (D) $\tan^{-1}2$

**MIT019**

**20.** $\lim_{n\to\infty}\sum_{r=1}^{n}\tan^{-1}\frac{2r+1}{r^4+2r^3+r^2+1}$ is equal to -

(A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $-\frac{\pi}{8}$

**MIT020**

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# EXERCISE (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

**1.** The values of '$x$' satisfying the equation $\sin^{-1}(x^2 - 5x + 7) = 2\tan^{-1}1$ is (are)

(A) 2 (B) 3 (C) 4 (D) 5

**MIT047**

**2.** Consider the function $f(x) = e^x$ and $g(x) = \sin^{-1} x$, then which of the following is/are necessarily true.

(A) Domain of $gof$ = Domain of $f$ (B) Range of $gof \subset$ Range of $g$

(C) Domain of $gof$ is $(-\infty, 0]$ (D) Range of $gof$ is $\left[-\frac{\pi}{2}, 0\right)$

**MIT048**

**3.** Let $f(x) = \sin^{-1}(tanx) + \cos^{-1}(cotx)$ then

(A) $f(x) = \frac{\pi}{2}$ wherever defined (B) domain of $f(x)$ is $x = n\pi \pm \frac{\pi}{4}, n \in I$

(C) period of $f(x)$is $\frac{\pi}{2}$ (D) $f(x)$ is many one function

**MIT049**

**4.** Which of the following is/are correct ?

(A) $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right) \forall\ x \in R - \{0\}$

(B) If $f : R \to R$ such that $f(x) = \text{sgn}(e^x)$ then $f(x)$ is an into function.

(C) If $f : R^+ \to R$ such that $f(x) = \sin x + x$ then $f(x)$ is an odd function.

(D) If $f : R \to R$ such that $f(x) = \frac{e^x}{e^{[x]}}$ then $f(x)$ is a periodic function.

(where [.] represents greatest integer function)

**MIT050**

**5.** If $\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + \cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) = \frac{p}{q}$, where $p$ & $q$ are relatively prime then digit at units place of $(p-q)^{2k+1}$, $k \in N$, can be -

(A) 1 (B) 3 (C) 7 (D) 9

**MIT051**

**6.** Value of $3\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ is greater than

(A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{5\pi}{6}$

**MIT052**

**7.** If $\alpha = 2\tan^{-1}(\sqrt{3-2\sqrt{2}}) + \sin^{-1}\left(\frac{1}{\sqrt{6}-\sqrt{2}}\right)$, $\beta = \cot^{-1}\left(\sqrt{3}-2\right) + \frac{1}{8}\sec^{-1}(-2)$ & $\gamma = \tan^{-1}\frac{1}{\sqrt{2}} + \cos^{-1}\frac{1}{\sqrt{3}}$, then

(A) $\alpha = \beta$ (B) $\alpha + \beta = 3\gamma$ (C) $4(\beta - \gamma) = \alpha$ (D) $\beta = \gamma$

**MIT053**

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**8.** If $\sin^{-1}\left(\frac{\sqrt{x}}{2}\right)+\sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right)+\tan^{-1}y=\frac{2\pi}{3}$, then:

(A) maximum value of $x^2+y^2$ is $\frac{49}{3}$ (B) maximum value of $x^2+y^2$ is 4

(C) minimum value of $x^2+y^2$ is $\frac{1}{3}$ (D) minimum value of $x^2+y^2$ is 3

**MIT054**

## COMPREHENSION TYPE QUESTIONS

### Paragraph for Question 9 to 11

Consider a continuous function such that each image has atmost three preimage & atleast one image has exactly three preimages. This type of function is to be called as three-one function.
On the basis of above information, answer the following questions :

**9.** Which of the following function is a three-one function?

(A) $|\ell n|x||$ (B) $e^{|x|}$ (C) $x^3+3x^2-7x+6$ (D) $\cos(\cos^{-1}x)$

**MIT055**

**10.** If $f(x)=\sin^{-1}(\sin x)$ is a three $-$ one function, then possible interval of $x$ is -

(A) $[-\pi,\pi]$ (B) $\left[-\frac{\pi}{2},\frac{3\pi}{2}\right]$ (C) $(-2\pi,0]$ (D) $\left[\frac{-3\pi}{2},\frac{\pi}{2}\right]$

**MIT056**

**11.** If $f(x)$ is a three-one function such that $f(a)=f(b)$ (where $a\neq b$), then number of maximum possible values of $b$ is -

(A) 1 (B) 2 (C) 3 (D) 4

**MIT057**

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### Paragraph for Question 12 to 13

If $\alpha=2\tan^{-1}\frac{1}{2}+\sin^{-1}\frac{3}{5}$, $\beta=\sin^{-1}\frac{12}{13}+\cos^{-1}\frac{4}{5}+\cot^{-1}\frac{16}{63}$.

On the basis of above information, answer the following questions :

**12.** If $2\sin\alpha$, $\cos\beta$ are roots of $x^2-ax+b=0$, then -

(A) $a=1$ (B) $a=-1$ (C) $b=-2$ (D) $\frac{b}{a}=-2$

**MIT058**

**13.** The range of $\cot^{-1}(x^2-2x)$ is $(c,d]$, then

(A) $c=0$ (B) $d=\frac{3\pi}{4}$ (C) $c=\frac{\pi}{4}$ (D) $c+d=\frac{3\pi}{4}$

**MIT059**

### Paragraph for Question 14 and 15

Let $\alpha=\sin^{-1}x-\cos^{-1}y$ & $\frac{\pi}{2}-\alpha=\sin^{-1}y+\cos^{-1}x$.

**14.** $\cos^{-1}(-y)$ is equal to -

(A) $\frac{3\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

**MIT060**

**15.** $[\alpha]$ can not be equal to (where [.] denotes greatest integer function)

(A) $-3$ (B) $-1$ (C) 0 (D) 1

**MIT061**



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## MATCHING LIST TYPE QUESTION

**Match List-I with List-II and select the correct answer.**

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | The number of integral values of $k$ for which the equation $\sin^{-1}x^2+\tan^{-1}x^2=2k+1$ has a solution | (1) | 1 |
| (Q) | Let $\sum_{k=1}^{\infty}\cot^{-1}\left(\frac{k^2}{8}\right)=\frac{p}{q}\pi$ where $\frac{p}{q}$ is rational in its lowest form, then find $\lvert q-p\rvert$ | (2) | 2 |
| (R) | If $\tan^{-1}\left(\sin^2\theta-2\sin\theta+3\right)+\cot^{-1}\left(5\sec^2 y+1\right)=\frac{\pi}{2}$, then the value of $\cos^2 y - \sin\theta$ | (3) | 3 |
| (S) | Minimum positive integral value of $x$ such that $f(x)$ is defined, $f(x)=\sqrt{\sec^{-1}\left(\frac{1-\lvert x\rvert}{2}\right)}$ | (4) | 0 |

**16.** Which of the following is the only **CORRECT** combination?

(A) (P), (3) (B) (S), (2) (C) (R), (2) (D) (Q), (1)

**MIT062**

**17.** Which of the following is the only **CORRECT** combination?

(A) (P), (1) (B) (Q), (1) (C) (R), (4) (D) (S), (2)

**MIT063**

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## MATRIX MATCH TYPE QUESTION

**18.** Match the column

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Let $a, b, c$ be three positive real numbers $\theta = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$, then $\theta$ is equal to | (p) | $\pi$ |
| (B) | The value of the expression $\tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}(\cot A) + \tan^{-1}(\cot^3 A)$ for $0 < A < (\pi/4)$ is equal to | (q) | $-\frac{\pi}{2}$ |
| (C) | If $x < 0$, then $\frac{1}{2}\{\cos-1(2x^2 - 1) + 2\cos-1 x\}$ is equal to | (r) | $-\pi$ |
| (D) | The value of $\sin^{-1}\left(\frac{3}{5}\right) - \cos^{-1}\left(\frac{12}{13}\right) + \cos^{-1}\left(\frac{16}{65}\right)$ is equal to | (s) | $\frac{\pi}{2}$ |

**MIT064** **MIT065** **MIT066** **MIT067**



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**19.** Match the column :

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (P) | $\tan^{-1}\left(\frac{3}{4}\right)+\tan^{-1}\left(\frac{3}{5}\right)-\tan^{-1}\left(\frac{8}{19}\right)$ | (1) | $\tan^{-1}\left(\frac{1}{2}\right)$ |
| (Q) | $\tan^{-1}\left(\frac{2}{11}\right)+\tan^{-1}\left(\frac{7}{24}\right)$ | (2) | $\frac{\pi}{4}$ |
| (R) | $2\cos^{-1}\frac{3}{\sqrt{13}}+\cot^{-1}\frac{16}{63}+\frac{1}{2}\cos^{-1}\frac{7}{25}$ | (3) | $\frac{\pi}{6}$ |
| (S) | $\text{arc cos}\sqrt{\frac{2}{3}}-arc\cos\frac{\sqrt{6}+1}{2\sqrt{3}}$ | (4) | $\pi$ |

**MIT087**

**20.** Solve for $x$ :

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (P) | $\sin^{-1}x+\sin^{-1}2x=\frac{\pi}{3}$ | (1) | $x=3$ |
| (Q) | $\tan^{-1}\frac{1}{1+2x}+\tan^{-1}\frac{1}{1+4x}=\tan^{-1}\frac{2}{x^2}$ | (2) | $x=\frac{1}{2}\sqrt{\frac{3}{7}}$ |
| (R) | $\tan^{-1}(x-1)+\tan^{-1}(x)+\tan^{-1}(x+1)=\tan^{-1}(3x)$ | (3) | $\left[\frac{\sqrt{3}}{2},1\right]$ |
| (S) | $3\cos^{-1}x=\sin^{-1}\left(\sqrt{1-x^2}(4x^2-1)\right)$ | (4) | $x=0,\frac{1}{2},-\frac{1}{2}$ |

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**MIT088**



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# EXERCISE (S)

**1.** The value of $\tan^2(\sec^{-1}3) + \cot^2(\text{cosec}^{-1}4)$ is –

**MIT022**

**2.** If $\sin^{-1}\sin\left(\frac{10\pi}{3}\right)+\cos^{-1}\cos\frac{22\pi}{3}+\tan^{-1}\tan 10 = a\pi + b$, then $\left|\frac{3ab}{80}\right|$ is equal to

**MIT030**

**3.** The value of $\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}}+\cos^{-1}\frac{\sqrt{12}}{4}+\sec^{-1}\sqrt{2}\right)\right\}$ is-

**MIT025**

**4.** If the value of $\cot\left(\text{cosec}^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is equal to $\frac{a}{b}$, where $a$ and $b$ are coprime, then the value of $a + b$ is

**MIT091**

**5.** Number of solution(s) of the equation $\cos^{-1}\sqrt{x}-\sin^{-1}\sqrt{x-1}+\cos^{-1}\sqrt{1-x}-\sin^{-1}\frac{1}{\sqrt{x}}=\frac{\pi}{2}$ is -

**MIT026**

**6.** The number of solution(s) of the equation, $\sin^{-1}x + \cos^{-1}(1 - x) = \sin^{-1}(-x)$, is

**MIT027**

**7.** Let $f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)+\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then the value of $f(1) + f(2)$ is $k\pi$ where $k$ is equal to

**MIT021**

**8.** If $x > 0$ $\cos^{-1}\left(\frac{12}{x}\right)=\frac{\pi}{2}-\cos^{-1}\left(\frac{16}{x}\right)$ then $x$ is -

**MIT023**

**9.** Number of integral ordered pairs $(a, b)$ for which

$\sin^{-1}(1 + b + b^2 + \ldots\infty) + \cos^{-1}\left(a-\frac{a^2}{3}+\frac{a^3}{9}-\ldots\infty\right)=\frac{\pi}{2}$ is –

**MIT024**

**10.** If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-\ldots\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-\ldots\right)=\frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then $x$ equals

**MIT028**

**11.** Let $y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \text{cosec}^{-1}(\text{cosec } 7)$. If $y$ simplifies to $a\pi + b$, then find $(a - b)$.

**MIT029**

**12.** If the domain of definition of the function $f(x)=\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ for real valued '$x$' is $[\alpha, \beta]$, then the value of $4(\alpha + \beta)$ is equal to:

**MIT068**

**13.** Number of solutions of equation $\tan^{-1}(e^{-x}) + \cot^{-1}(|\ell nx|) = \pi/2$ is :

**MIT069**

**14.** If $\tan^{-1}x + \cot^{-1}\frac{1}{y} + 2\tan^{-1}z = \pi$, then $x + y + \mu z = xz^2 + yz^2 + \lambda xyz$. Find $\lambda + \mu$

**MIT089**

**15.** If $a > 0, b > 0, \tan^{-1}\left(\frac{a}{x}\right)+\tan^{-1}\left(\frac{b}{x}\right)=\frac{\pi}{2}$ then $x=(ab)^{\frac{1}{n}}$ find $n(n \in N)$ **MIT090**

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# EXERCISE - JEE (Main) PYQ

**1.** If $\cos^{-1}\left(\frac{2}{3x}\right)+\cos^{-1}\left(\frac{3}{4x}\right)=\frac{\pi}{2}, \left(x>\frac{3}{4}\right)$, then $x$ is equal to : **[JEE (Main) 2019]**

(1) $\frac{\sqrt{145}}{12}$ (2) $\frac{\sqrt{145}}{10}$ (3) $\frac{\sqrt{146}}{12}$ (4) $\frac{\sqrt{145}}{11}$

**MIT031**

**2.** If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to: **[JEE (Main) 2019]**

(1) $\pi$ (2) $7\pi$ (3) 0 (4) 10

**MIT032**

**3.** The value of $\cot\left(\sum_{n=1}^{19}\cot^{-1}\left(1+\sum_{p=1}^{n}2p\right)\right)$ is : **[JEE (Main) 2019]**

(1) $\frac{22}{23}$ (2) $\frac{23}{22}$ (3) $\frac{21}{19}$ (4) $\frac{19}{21}$

**MIT033**

**4.** Considering only the principal values of inverse functions, the set

$A=\left\{x\geq 0:\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}\right\}$ **[JEE (Main) 2019]**

(1) is an empty set
(2) Contains more than two elements
(3) Contains two elements
(4) is a singleton

**MIT034**

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**5.** The value of $\sin^{-1}\left(\frac{12}{13}\right)-\sin^{-1}\left(\frac{3}{5}\right)$ is equal to : **[JEE (Main) 2019]**

(1) $\pi-\sin^{-1}\left(\frac{63}{65}\right)$ (2) $\pi-\cos^{-1}\left(\frac{33}{65}\right)$ (3) $\frac{\pi}{2}-\sin^{-1}\left(\frac{56}{65}\right)$ (4) $\frac{\pi}{2}-\cos^{-1}\left(\frac{9}{65}\right)$

**MIT037**

**6.** If $S$ is the sum of the first 10 terms of the series $\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\tan^{-1}\left(\frac{1}{21}\right)+\ldots$,

then $\tan(S)$ is equal to : **[JEE (Main) 2020]**

(1) $\frac{5}{11}$ (2) $-\frac{6}{5}$ (3) $\frac{10}{11}$ (4) $\frac{5}{6}$

**MIT038**

**7.** The domain of the function $f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$ is $(-\infty, -a]\cup[a, \infty)$. Then a is equal to :

**[JEE (Main) 2020]**

(1) $\frac{1+\sqrt{17}}{2}$ (2) $\frac{\sqrt{17}-1}{2}$ (3) $\frac{\sqrt{17}}{2}+1$ (4) $\frac{\sqrt{17}}{2}$

**MIT092**

**8.** $2\pi-\left(\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{16}{65}\right)$ is equal to: **[JEE (Main) 2020]**

(1) $\frac{7\pi}{4}$ (2) $\frac{5\pi}{4}$ (3) $\frac{3\pi}{2}$ (4) $\frac{\pi}{2}$

**MIT093**



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**9.** Let $M$ and $m$ respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $\left[0, \frac{\pi}{2}\right]$. Then the value of $\tan(M - m)$ is equal to: **[JEE (Main) 2021]**

(1) $2+\sqrt{3}$ (2) $2-\sqrt{3}$ (3) $3+2\sqrt{2}$ (4) $3-2\sqrt{2}$

**MIT039**

**10.** If $\sum_{r=1}^{50} \tan^{-1}\frac{1}{2r^2} = p$, then the value of $\tan p$ is: **[JEE (Main) 2021]**

(1) $\frac{101}{102}$ (2) $\frac{50}{51}$ (3) 100 (4) $\frac{51}{50}$

**MIT040**

**11.** If $0 < a, b < 1$, and $\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$, then the value of $(a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \ldots$ is : **[JEE (Main) 2021]**

(1) $\log_e 2$ (2) $e^2 - 1$ (3) $e$ (4) $\log_e\left(\frac{e}{2}\right)$

**MIT041**

**12.** The domain of the function $f(x) = \sin^{-1}\left(\frac{3x^2+x-1}{(x-1)^2}\right) + \cos^{-1}\left(\frac{x-1}{x+1}\right)$ is : **[JEE (Main) 2021]**

(1) $\left[0, \frac{1}{4}\right]$ (2) $[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2}\right]$ (3) $\left[\frac{1}{4}, \frac{1}{2}\right] \cup \{0\}$ (4) $\left[0, \frac{1}{2}\right]$

**MIT094**

**13.** $\operatorname{cosec}\left[2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to: **[JEE (Main) 2021]**

(1) $\frac{56}{33}$ (2) $\frac{65}{56}$ (3) $\frac{65}{33}$ (4) $\frac{75}{56}$

**MIT095**

**14.** The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$ is : **[JEE (Main) 2022]**

(1) $R - \left\{-\frac{1}{2}, \frac{1}{2}\right\}$ (2) $(-\infty, -1] \cup [1, \infty) \cup \{0\}$

(3) $\left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \cup \{0\}$ (4) $\left(-\infty, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

**MIT042**

**15.** The set of all values of $k$ for which $(\tan^{-1}x)^3 + (\cot^{-1}x)^3 = k\pi^3, x \in R$, is the interval: **[JEE (Main) 2022]**

(1) $\left[\frac{1}{32}, \frac{7}{8}\right)$ (2) $\left(\frac{1}{24}, \frac{13}{16}\right)$ (3) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (4) $\left[\frac{1}{32}, \frac{9}{8}\right)$

**MIT043**

**16.** If the inverse trigonometric functions take principal values, then $\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + \frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to: **[JEE (Main) 2022]**

(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{6}$

**MIT045**



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**17.** The domain of the function $f(x)=\frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)}$ is **[JEE (Main) 2022]**

(1) $(-\infty,1)\cup(2,\infty)$ (2) $(2,\infty)$

(3) $\left[-\frac{1}{2},1\right)\cup(2,\infty)$ (4) $\left[-\frac{1}{2},1\right)\cup(2,\infty)-\left\{\frac{3+\sqrt{5}}{2},\frac{3-\sqrt{5}}{2}\right\}$

**MIT046**

**18.** Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$. If $[a, b]$ is the range of the function then $4a - b$ is equal to: **[JEE (Main) 2022]**

(1) 11 (2) $11 - \pi$ (3) $11 + \pi$ (4) $15 - \pi$

**MIT096**

**19.** $50\ tan\left(3tan^{-1}\left(\frac{1}{2}\right)+2cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right)+4\sqrt{2}\tan\left(\frac{1}{2}tan^{-1}(2\sqrt{2})\right)$ is equal to. **[JEE (Main) 2022]**

**MIT097**

**20.** The range of $f(x)=4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is **[JEE (Main) 2023]**

(1) $[0, \pi]$ (2) $[0, 2\pi)$ (3) $[0, \pi)$ (4) $[0, 2\pi]$

**MIT081**

**21.** $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right)+\sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to **[JEE (Main) 2023]**

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{6}$

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**MIT082**

**22.** If the sum of all the solutions of $\tan^{-1}\left(\frac{2x}{1-x^2}\right)+\cot^{-1}\left(\frac{1-x^2}{2x}\right)=\frac{\pi}{3}$, $-1<x<1, x\neq 0$, is $\alpha-\frac{4}{\sqrt{3}}$, then $\alpha$ is equal to. **[JEE (Main) 2023]**

**MIT083**

**23.** Let $a_1=1,a_2,a_3,a_4,.....$ be consecutive natural numbers. Then $\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)$ $+.....+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$ is equal to **[JEE (Main) 2023]**

(1) $\frac{\pi}{4}-\cot^{-1}(2022)$ (2) $\cot^{-1}(2022)-\frac{\pi}{4}$ (3) $\tan^{-1}(2022)-\frac{\pi}{4}$ (4) $\frac{\pi}{4}-\tan^{-1}(2022)$

**MIT084**

**24.** Let $(a, b) \subset (0, 2\pi)$ be the largest interval for which $\sin^{-1}(\sin\theta)-\cos^{-1}(\sin\theta)>0, \theta\in(0,2\pi)$, holds. If $\alpha x^2+\beta x+\sin^{-1}\left(x^2-6x+10\right)+\cos^{-1}\left(x^2-6x+10\right)=0$ and $\alpha-\beta=b-a$, then $\alpha$ is equal to : **[JEE (Main) 2023]**

(1) $\frac{\pi}{48}$ (2) $\frac{\pi}{16}$ (3) $\frac{\pi}{8}$ (4) $\frac{\pi}{12}$

**MIT085**



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**25.** If $\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}=0$, $0<\alpha<13$, then $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)$ is equal to

(1) $\pi$ (2) 16 (3) 0 (4) $16-5\pi$

**[JEE (Main) 2023]**

**MIT086**

**26.** Given the inverse trigonometric function assumes principal values only. Let $x, y$ be any two real numbers in $[-1,1]$ such that $\cos^{-1}x - \sin^{-1}y = \alpha, \frac{-\pi}{2}\le\alpha\le\pi$. Then, the minimum value of $x^2 + y^2 + 2xy\sin\alpha$ is **[JEE (Main) 2024]**

(1) $-1$ (2) 0 (3) $\frac{-1}{2}$ (4) $\frac{1}{2}$

**MIT148**

**27.** For $n \in N$, if $\cot^{-1}3 + \cot^{-1}4 + \cot^{-1}5 + \cot^{1} n=\frac{\pi}{4}$, then $n$ is equal to _____. **[JEE (Main) 2024]**

**MIT149**

**28.** If the domain of the function $f(x)=\sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $R-(\alpha,\beta)$ then $12\alpha\beta$ is equal to:

**[JEE (Main) 2024]**

(1) 36 (2) 24 (3) 40 (4) 32

**MIT150**

**29.** Let $x=\frac{m}{n}$ ($m, n$ are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1}x\right)=\frac{1}{9}$ and let $\alpha, \beta(\alpha>\beta)$ be the roots of the equation $mx^2 - nx - m + n = 0$. Then the point $(\alpha,\beta)$ lies on the line **[JEE (Main) 2024]**

(1) $3x + 2y = 2$ (2) $5x - 8y = -9$ (3) $3x - 2y = -2$ (4) $5x + 8y = 9$

**MIT151**

**30.** If $a=\sin^{-1}(\sin(5))$ and $b=\cos^{-1}(\cos(5))$, then $a^2+b^2$ is equal to **[JEE (Main) 2024]**

(1) $4\pi^2+25$ (2) $8\pi^2-40\pi+50$ (3) $4\pi^2-20\pi+50$ (4) 25

**MIT152**

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# EXERCISE - JEE (Advanced) PYQ

**1.** The value of $\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$ is **[JEE (Advanced) 2013]**

(A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

**MIT070**

**2.** Match List-I with List-II and select the correct answer using the code given below the lists.

| | **List-I** | | **List-II** |
|---|---|---|---|
| P. | $\left(\frac{1}{y^2}\left(\frac{\cos\left(\tan^{-1}y\right)+y\sin\left(\tan^{-1}y\right)}{\cot\left(\sin^{-1}y\right)+\tan\left(\sin^{-1}y\right)}\right)^2+y^4\right)^{1/2}$ takes value | 1. | $\frac{1}{2}\sqrt{\frac{5}{3}}$ |
| Q. | If $\cos x+\cos y+\cos z=0=\sin x+\sin y+\sin z$, then possible value of $\cos\frac{x-y}{2}$ is | 2. | $\sqrt{2}$ |
| R. | If $\cos\left(\frac{\pi}{4}-x\right)\cos 2x+\sin x\sin 2x\sec x$ $=\cos x\sin 2x\sec x+\cos\left(\frac{\pi}{4}+x\right)\cos 2x$, then possible value of $\sec x$ is | 3. | $\frac{1}{2}$ |
| S. | If $\cot\left(\sin^{-1}\sqrt{1-x^2}\right)=\sin\left(\tan^{-1}\left(x\sqrt{6}\right)\right), x\neq 0$, then possible value of $x$ is | 4. | 1 |

**[JEE (Advanced) 2013]**

**Codes :**

| | **P** | **Q** | **R** | **S** |
|---|---|---|---|---|
| (A) | 4 | 3 | 1 | 2 |
| (B) | 4 | 3 | 2 | 1 |
| (C) | 3 | 4 | 2 | 1 |
| (D) | 3 | 4 | 1 | 2 |

**MIT071**

**3.** Let $f:[0,4\pi]\to[0,\pi]$ be defined by $f(x)=\cos^{-1}(\cos x)$. The number of points $x\in[0,4\pi]$ satisfying the equation $f(x)=\frac{10-x}{10}$ is **[JEE (Advanced) 2014]**

**MIT072**

**4.** If $\alpha=3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta=3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct option(s) is(are) **[JEE (Advanced) 2015]**

(A) $\cos\beta>0$ (B) $\sin\beta<0$ (C) $\cos(\alpha+\beta)>0$ (D) $\cos\alpha<0$

**MIT073**

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**5.** The number of real solutions of the equation

$\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty}(-x)^i\right)$ lying in the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$ is ___

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume value in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.)

**[JEE (Advanced) 2018]**

**MIT074**

**6.** Let $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0\right\}$ and $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ is a real number}\right\}$.

$\left(\text{Here, the inverse trigonometric function } \sin^{-1}x \text{ assumes values in } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].\right)$

Let $f : E_1 \to \mathbb{R}$ be the function defined by $f(x) = \log_e\left(\frac{x}{x-1}\right)$

and g : $E_2 \to \mathbb{R}$ be the function defined by $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$.

| | LIST-I | | LIST-II |
|---|---|---|---|
| **P.** | The range of $f$ is | **1.** | $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$ |
| **Q.** | The range of $g$ contains | **2.** | $(0, 1)$ |
| **R.** | The domain of $f$ contains | **3.** | $\left[-\frac{1}{2}, \frac{1}{2}\right]$ |
| **S.** | The domain of $g$ is | **4.** | $(-\infty, 0) \cup (0, \infty)$ |
| | | **5.** | $\left(-\infty, \frac{e}{e-1}\right]$ |
| | | **6.** | $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1}\right]$ |

The correct option is : **[JEE (Advanced) 2018]**

(A) P $\to$ 4; Q $\to$ 2; R $\to$ 1; S $\to$ 1 (B) P $\to$ 3; Q $\to$ 3; R $\to$ 6; S $\to$ 5

(C) P $\to$ 4; Q $\to$ 2; R $\to$ 1; S $\to$ 6 (D) P $\to$ 4; Q $\to$ 3; R $\to$ 6; S $\to$ 5

**MIT075**

**7.** The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12} + \frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12} + \frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals

**[JEE (Advanced) 2019]**

**MIT076**



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**8.** For any positive integer $n$, let $S_n : (0, \infty) \to \mathbb{R}$ be defined by $S_n(x) = \sum_{k=1}^{n} \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$, where for any $x \in \mathbb{R}$, $\cot^{-1}x \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) **TRUE** ? **[JEE (Advanced) 2021]**

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x > 0$

(B) $\lim_{n\to\infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \le \frac{1}{2}$, for all $n \ge 1$ and $x > 0$

**MIT077**

**9.** Considering only the principal values of the inverse trigonometric functions, the value of $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}} + \frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2} + \tan^{-1}\frac{\sqrt{2}}{\pi}$ is _____. **[JEE (Advanced) 2022]**

**MIT078**

**10.** Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in \mathbb{R}$. Then the number of real solutions of the equation $\sqrt{1+\cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to

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**[JEE (Advanced) 2023]**

**MIT079**

**11.** For any $y \in \mathbb{R}$, let $\cot^{-1}(y) \in (0, \pi)$ and $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the sum of all the solutions of the equation $\tan^{-1}\left(\frac{6y}{9-y^2}\right) + \cot^{-1}\left(\frac{9-y^2}{6y}\right) = \frac{2\pi}{3}$ for $0 < |y| < 3$, is equal to

**[JEE (Advanced) 2023]**

(A) $2\sqrt{3} - 3$ (B) $3 - 2\sqrt{3}$ (C) $4\sqrt{3} - 6$ (D) $6 - 4\sqrt{3}$

**MIT080**

**12.** Considering only the principal values of the inverse trigonometric functions, the value of $\tan\left(\sin^{-1}\left(\frac{3}{5}\right) - 2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ is **[JEE (Advanced) 2024]**

(A) $\frac{7}{24}$ (B) $\frac{-7}{24}$ (C) $\frac{-5}{24}$ (D) $\frac{5}{24}$

**MIT153**



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# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | C | B | A | C | C | C | D | B | A | B |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A | B | B | A | C | C | A | A | B | A |

## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A,B | B,C | A,B,C,D | B,D | B,C | A,B | A,C | A,C | C | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | | |
| Ans. | B | A,C,D | A,B,D | A | D | C | A | A → p ; B → p ; C → p ; D → s | | |
| Que. | 19 | | | | 20 | | | | | |
| Ans. | P → 2; Q → 1; R → 4; S → 3 | | | | P → 2; Q → 1; R → 4; S → 3 | | | | | |

## EXERCISE (S)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** | 23 | **2.** | 1 | **3.** | 0 | **4.** | 23 | **5.** | 0 |
| **6.** | 1 | **7.** | 2 | **8.** | 20 | **9.** | 0 | **10.** | 1 |
| **11.** | 53 | **12.** | 1 | **13.** | 2 | **14.** | 4 | **15.** | 2 |

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## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 1 | 1 | 3 | 4 | 3 | 4 | 1 | 3 | 4 | 2 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 1 | 3 | 2 | 4 | 1 | 3 | 4 | 2 | 29 | 2 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | 3 | 2 | 1,3 | 4 | 1 | 2 | 47 | 4 | 4 | 2 |

## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | B | 3 | B,C,D | 2 | A | 0.00 | A,B | 2.35 or 2.36 | 3 |
| Que. | 11 | 12 | | | | | | | | |
| Ans. | C | B | | | | | | | | |



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